Day 11 — Networking Fundamentals

1. Introduction

Computer networking is the process of connecting two or more devices to share information, resources, and services. Networks allow computers, servers, smartphones, and other devices to communicate with each other.

Networking fundamentals are important in cybersecurity because they help security professionals understand how data travels, how devices communicate, and how attackers may exploit network weaknesses.

2. IP Address

An IP (Internet Protocol) address is a logical address assigned to a device on a network. It helps identify the device and enables data to be routed to the correct destination.

There are two main versions of IP addresses:

- IPv4: Uses a 32-bit address, usually written as four decimal numbers separated by dots. Example: 192.168.1.10
- IPv6: Uses a 128-bit address written in hexadecimal notation. Example: 2001:db8::1

IP addresses can also be classified as public or private. Public IP addresses are used for communication over the internet, while private IP addresses are commonly used within local networks.

3. MAC Address

A MAC (Media Access Control) address is a link-layer address used to identify a network interface on a local network. It is commonly represented as six pairs of hexadecimal digits.

Example: 00:1A:2B:3C:4D:5E

MAC addresses help switches forward Ethernet frames within a local network. Unlike IP addresses, which support logical addressing and routing, MAC addresses are primarily used for communication on the local network link.

4. Network Ports

A network port is a logical number used by transport-layer protocols to direct network traffic to the appropriate application or service on a device.

Common examples include:

- Port 80: HTTP web traffic.
- Port 443: HTTPS secure web traffic.
- Port 22: SSH remote administration.
- Port 25: SMTP email transfer.
- Port 53: DNS queries, commonly using UDP or TCP.

Ports help multiple applications communicate over a network. In cybersecurity, unnecessary exposed ports can increase risk, so organizations should restrict access to services that are not required.

5. Network Protocols

Network protocols are rules that define how devices exchange information.

TCP (Transmission Control Protocol): Provides reliable, ordered delivery of data between applications.

UDP (User Datagram Protocol): Sends data with lower overhead but does not guarantee delivery or ordering.

HTTP and HTTPS: Used for communication between web browsers and web servers. HTTPS protects communication using TLS.

DNS (Domain Name System): Translates domain names, such as example.com, into IP addresses and other records.

DHCP (Dynamic Host Configuration Protocol): Automatically provides devices with network configuration, including IP addresses.

IP (Internet Protocol): Provides addressing and routing so packets can travel across interconnected networks.

6. Network Devices

Router

A router connects different IP networks and forwards packets toward their destinations. Home routers commonly connect a local network to an internet service provider.

Switch

A network switch connects devices within a local area network (LAN). It learns MAC addresses and uses them to forward Ethernet frames to the appropriate ports.

Firewall

A firewall filters network traffic according to configured security rules. It can allow legitimate communications and block unauthorized or unwanted connections.

Modem

A modem connects a network to an internet service provider through the provider's access technology.

Access Point

A wireless access point allows Wi-Fi devices to connect to a network.

7. Importance of Networking in Cybersecurity

Networking knowledge helps cybersecurity professionals understand network traffic, identify suspicious connections, and investigate security incidents. IP addresses can help identify communication endpoints, ports reveal which services may be exposed, and protocols explain how data is exchanged.

Routers, switches, and firewalls also play important roles in network segmentation, traffic management, access control, and threat prevention.

Networking fundamentals provide the foundation for understanding communication between devices. IP addresses identify network endpoints, MAC addresses support local-link communication, ports direct traffic to applications, and protocols define how information is exchanged.

Understanding routers, switches, and firewalls is essential for recognizing network vulnerabilities and implementing effective cybersecurity controls.